

Operating System Security

Lab 10

Using UFW and SSH

Alex MacIntyre

Requirements:

Installing and configuring the Ubuntu Firewall

- **Install the firewall & confirmed it is installed (Yes you may already have it installed please show how to do the process)**
- **Show the firewall is enabled and running**
- **Show how to see what ports are open and what ports are listening**

Using SSH

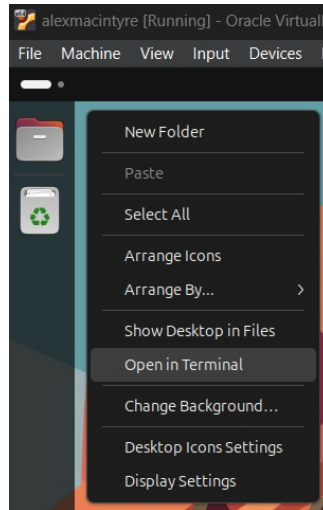
- **Install SSH:**
- **Install PUTTY:**
- **Create the connection:**
- **Once complete, go change the ssh port in the config file**
- **Use Putty to show the connection now working with the new ssh port**
- **Change the ssh port back port 22**

Scripting: Passing Command Arguments/Parameters to a Script

- **Create a script and then run it to pass the parameters to it**
- **IPV4**
- **Subnet Mask**
- **Default Gateway**
- **Then display all three parameters on the same line: "Config updates are:"**
- **Show the Script Code**
- **Run the Script and show the results passing the configurations**

Part I – Installing and Configuring the Ubuntu Firewall

Step One – Open the terminal. This can be done by pressing “CTRL + ALT + T” on the keyboard, or by right clicking the desktop and pressing “Open in Terminal”.



Step Two - Once the terminal is open, ensure that Ubuntu is fully updated and upgraded.

To update Ubuntu, type “**sudo apt update**” into the terminal and press **Enter**.

To upgrade Ubuntu, type “**sudo apt upgrade**” and press **Enter**. There will be a prompt to complete the operation. Type “**Y**” and press **Enter**.

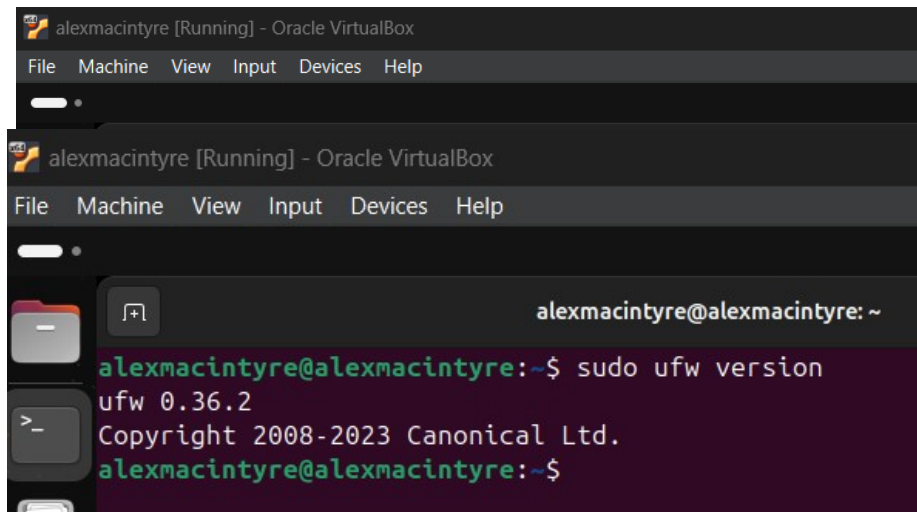
Two side-by-side terminal window screenshots from an Oracle VM VirtualBox. The left terminal shows the output of 'sudo apt update', listing updates for various Ubuntu components like 'noble-security', 'noble-backports', and 'noble-updates'. The right terminal shows the output of 'sudo apt upgrade', indicating that several packages will be upgraded, including 'libcodecs', 'libcodecs', 'libcodecs', 'libcodecs', 'libcodecs', and 'libcodecs'. It also shows the disk space requirements for the upgrade.

```
alexmacintyre@alexmacintyre:~$ sudo apt update
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ca.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [673 kB]
Get:6 http://ca.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [158 kB]
Get:7 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [131 kB]
Get:8 http://ca.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:9 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [719 kB]
Get:10 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [214 kB]
Get:11 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [310 kB]
Get:12 http://ca.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://ca.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [208 B]
Get:14 http://ca.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Co
```

```
alexmacintyre@alexmacintyre:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
libcodecs1 libcodecs60 libavutil58 libswscale7 libswresample4 libavformat60
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following upgrades have been deferred due to phasing:
python3-distupgrader ubuntu-release-upgrader-core
ubuntu-release-upgrader-gtk
The following packages will be upgraded:
dmdcode gir1.2-soup-3.0 libsoup-2.4-1 libsoup-3.0-0 libsoup-3.0-common
libsoup2.4-common snapd vin-common vin-tiny xxd
10 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
8 standard LTS security updates
Need to get 32.0 MB of archives.
After this operation, 3,430 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Step Three – Install UFW by typing “**sudo apt install ufw**” and pressing **Enter**.

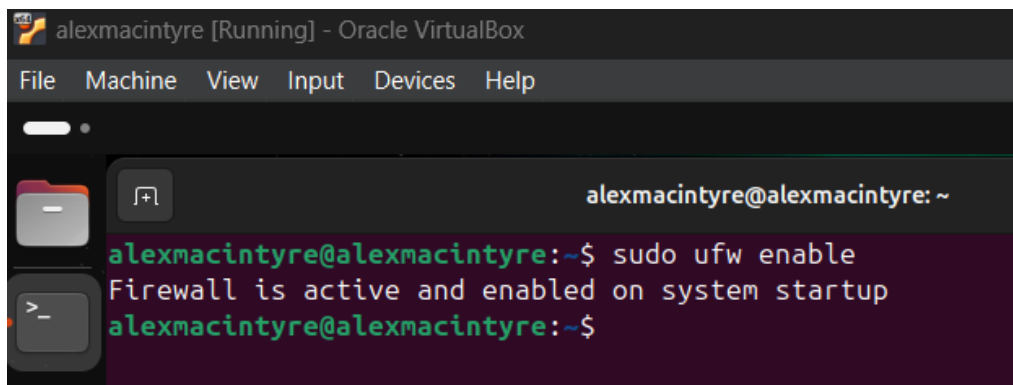
Step Four – Confirm that UFW was successfully installed by typing “**sudo ufw version**”.



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo ufw version
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.
alexmacintyre@alexmacintyre:~$
```

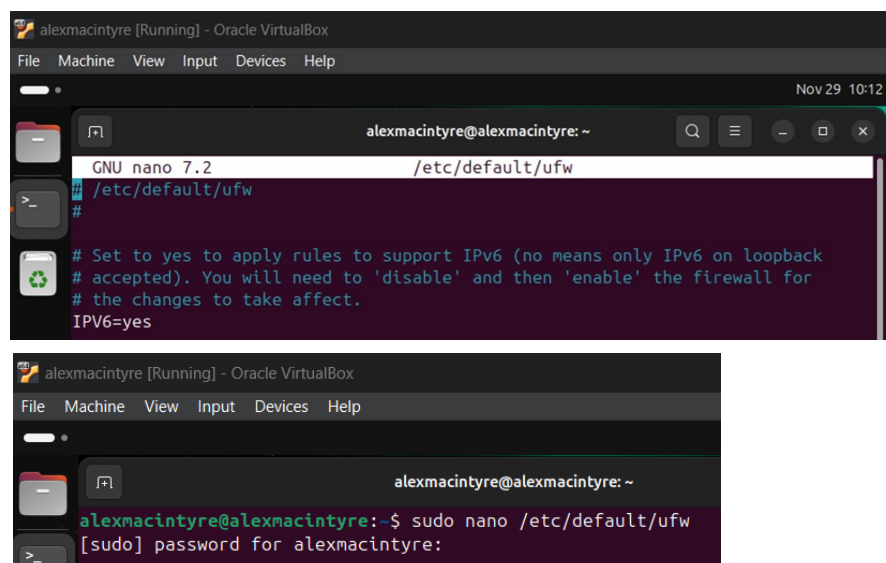
Step Five – Enable the firewall by typing “**sudo ufw enable**” and pressing **Enter**.



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo ufw enable
Firewall is active and enabled on system startup
alexmacintyre@alexmacintyre:~$
```

Step Six – To verify the IPv6 status, type “**sudo nano /etc/default/ufw**” and press **Enter** to open the Text Editor. Enter the **sudo password** and press **Enter**. When the Text Editor is open, IPv6 should say “**yes**” next to it. If it doesn’t, change it to “**yes**” and press “**CTRL+X**” to save and exit.



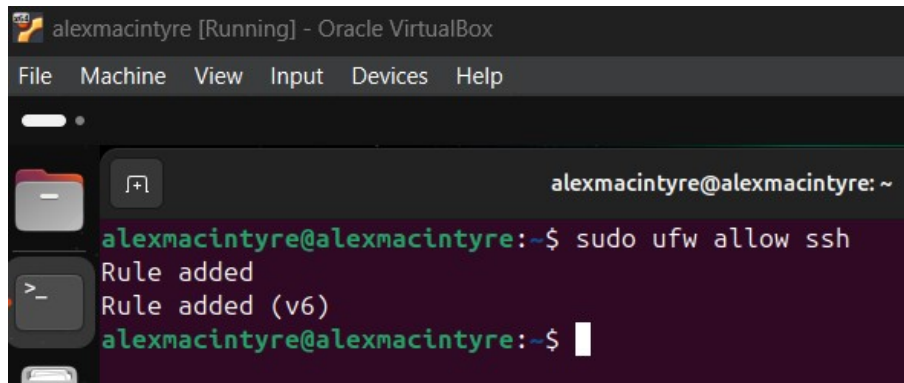
```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Nov 29 10:12
alexmacintyre@alexmacintyre: ~
GNU nano 7.2 /etc/default/ufw
#
# Set to yes to apply rules to support IPv6 (no means only IPv6 on loopback
# accepted). You will need to 'disable' and then 'enable' the firewall for
# the changes to take affect.
IPv6=yes

alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo nano /etc/default/ufw
[sudo] password for alexmacintyre:
```

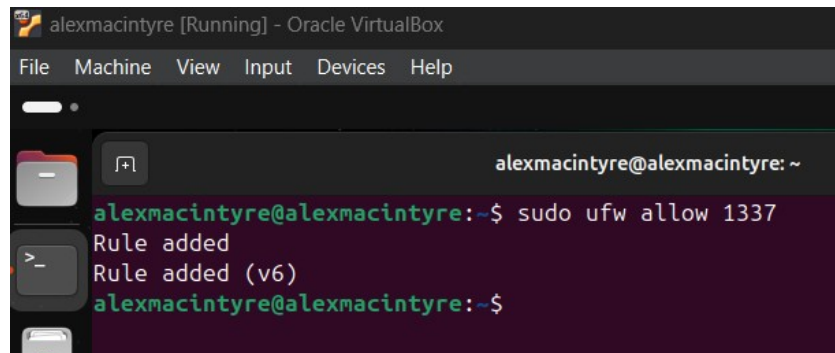
Step Seven – Add a rule to the firewall that allows SSH. Type “`sudo ufw allow ssh`” and press **Enter**.



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo ufw allow ssh
Rule added
Rule added (v6)
alexmacintyre@alexmacintyre:~$
```

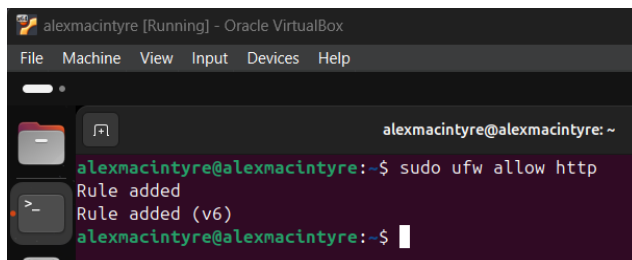
Step Eight – Add a rule to the firewall that enables port number 1337.



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

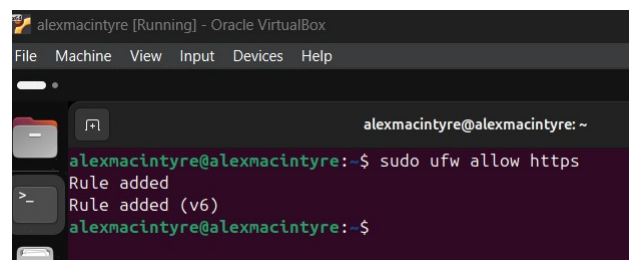
alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo ufw allow 1337
Rule added
Rule added (v6)
alexmacintyre@alexmacintyre:~$
```

Step Nine - Add a rule to the firewall that allows HTTP and HTTPS connections. For HTTP, type “`sudo ufw allow http`” and press **Enter**. For HTTPS, type “`sudo ufw allow https`” and press **Enter**.



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

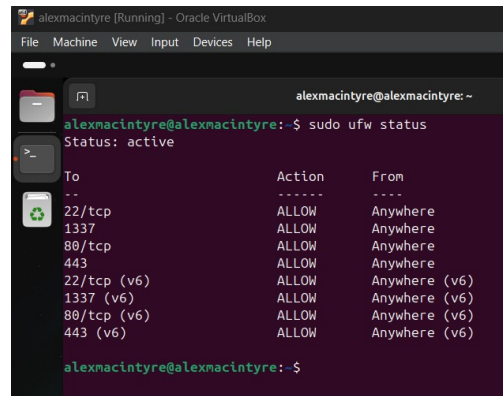
alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo ufw allow http
Rule added
Rule added (v6)
alexmacintyre@alexmacintyre:~$
```



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help

alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo ufw allow https
Rule added
Rule added (v6)
alexmacintyre@alexmacintyre:~$
```

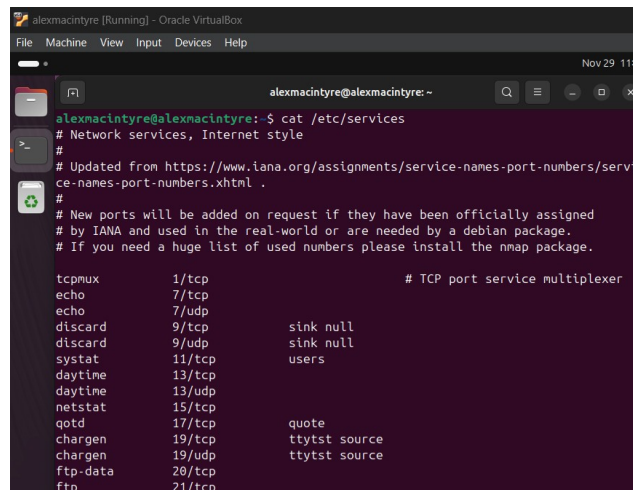
Step Ten – To check the status of the firewall and rule configurations, type “**sudo ufw status**” and press **Enter**.



```
alexmacintyre@alexmacintyre:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
1337 ALLOW Anywhere
80/tcp ALLOW Anywhere
443 ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
1337 (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)
```

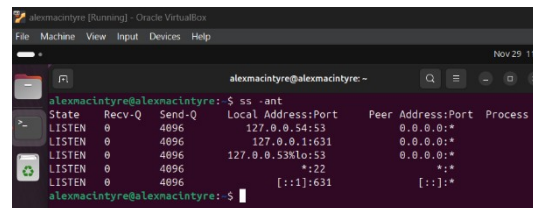
Step Eleven – To view **Network Services**, which includes a list of ports that are in use, type “**cat /etc/services**” and press **Enter**.



```
alexmacintyre@alexmacintyre:~$ cat /etc/services
# Network services, Internet style
#
# Updated from https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml
#
# New ports will be added on request if they have been officially assigned
# by IANA and used in the real-world or are needed by a debian package.
# If you need a huge list of used numbers please install the nmap package.

tcpmux      1/tcp          # TCP port service multiplexer
echo        7/tcp
echo        7/udp
discard     9/tcp          sink null
discard     9/udp          sink null
systat      11/tcp         users
daytime     13/tcp
daytime     13/udp
netstat     15/tcp
qotd        17/tcp          quote
chargen     19/tcp          ttytst source
chargen     19/udp          ttytst source
ftp-data    20/tcp
ftp         21/tcp
```

Step Twelve – To see which ports are listening, type “**ss -ant**” and press **Enter**.

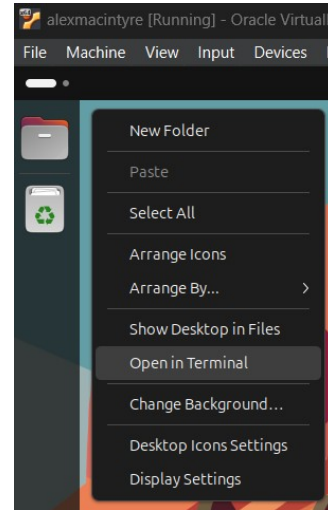


```
alexmacintyre@alexmacintyre:~$ ss -ant
State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
LISTEN 0 4096 127.0.0.1:631 0.0.0.0:*
LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
LISTEN 0 4096 *:22 *:*
LISTEN 0 4096 [::]:631 [::]:*
```

Part II – Using SSH

Installing OpenSSH on Linux

Step One – Open the terminal. This can be done by pressing “CTRL + ALT + T” on the keyboard, or by right clicking the desktop and pressing “Open in Terminal”.



Step Two – Once the terminal is open, ensure that Ubuntu is fully updated and upgraded.

To update Ubuntu, type “**sudo apt update**” into the terminal and press **Enter**.

To upgrade Ubuntu, type “**sudo apt upgrade**” and press **Enter**. There will be a prompt to complete the operation. Type “**Y**” and press **Enter**.

Three screenshots of a terminal window in an Oracle VM VirtualBox environment, showing the process of updating and upgrading Ubuntu and installing OpenSSH. The first screenshot shows the command 'sudo apt update' being executed, with output showing the progress of updating package lists. The second screenshot shows the command 'sudo apt upgrade' being executed, with output showing the progress of upgrading packages. The third screenshot shows the command 'sudo apt install openssh-server' being executed, with output showing the progress of installing the OpenSSH server package.

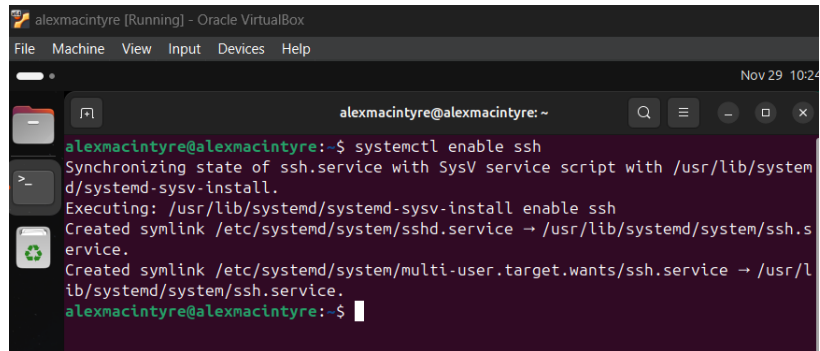
```
alexmacintyre@alexmacintyre:~$ sudo apt update
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ca.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [673 kB]
Get:6 http://ca.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [158 kB]
Get:7 http://ca.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [131 kB]
Get:8 http://ca.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:9 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [719 kB]
Get:10 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [214 kB]
Get:11 http://ca.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [310 kB]
Get:12 http://ca.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://ca.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [208 B]
Get:14 http://ca.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Co
```

```
alexmacintyre@alexmacintyre:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
libbson1 libavcodec60 libavutil58 libswscale7 libswresample4 libavformat60
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following upgrades have been deferred due to phasing:
python3-distupgrader ubuntu-release-upgrader-core
ubuntu-release-upgrader-gtk
The following packages will be upgraded:
dmdcode gir1.2-soup-3.0 libsoup-2.4-1 libsoup-3.0-0 libsoup-3.0-common
libsoup2.4-common snand vim-common vim-tiny xxd
```

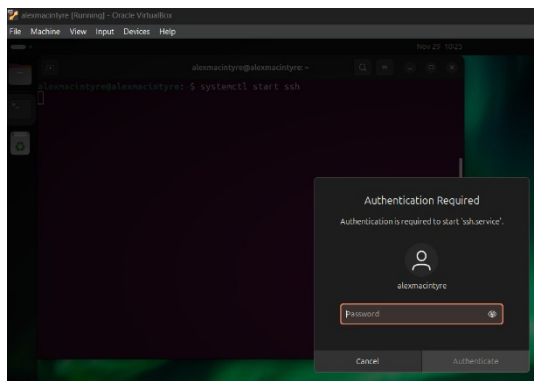
```
alexmacintyre@alexmacintyre:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 3 not upgraded.
Need to get 832 kB of archives.
After this operation, 6,747 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Step Three – Install OpenSSH by typing “**sudo apt install openssh-server**” and pressing **Enter**. There will be a prompt to continue. To continue, type “**Y**” and press **Enter**.

Step Four – Enable SSH by typing “**systemctl enable ssh**” and pressing **Enter**. A dialogue will appear asking for authentication. Enter the sudo password and click “**Authenticate**” repeatedly until the dialogue disappears.

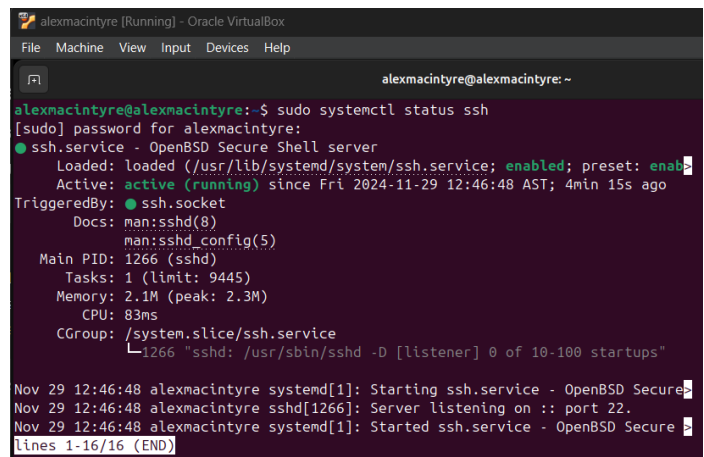


```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Nov 29 10:24
alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
alexmacintyre@alexmacintyre:~$
```



Step Five – Start SSH by typing “**systemctl start ssh**” and press **Enter**. A dialogue will appear asking for authentication. Enter the sudo password and click “**Authenticate.**”

Step Six – To view the status of SSH, type “**sudo systemctl status ssh**” and press Enter. It should say “**active (running)**”.



```
alexmacintyre [Running] - Oracle VirtualBox
File Machine View Input Devices Help
alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo systemctl status ssh
[sudo] password for alexmacintyre:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Fri 2024-11-29 12:46:48 AST; 4min 15s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1266 (sshd)
      Tasks: 1 (limit: 9445)
     Memory: 2.1M (peak: 2.3M)
        CPU: 83ms
    CGroup: /system.slice/ssh.service
            └─1266 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Nov 29 12:46:48 alexmacintyre systemd[1]: Starting ssh.service - OpenBSD Secure Shell server:
Nov 29 12:46:48 alexmacintyre sshd[1266]: Server listening on :: port 22.
Nov 29 12:46:48 alexmacintyre systemd[1]: Started ssh.service - OpenBSD Secure Shell server:
lines 1-16/16 (END)
```

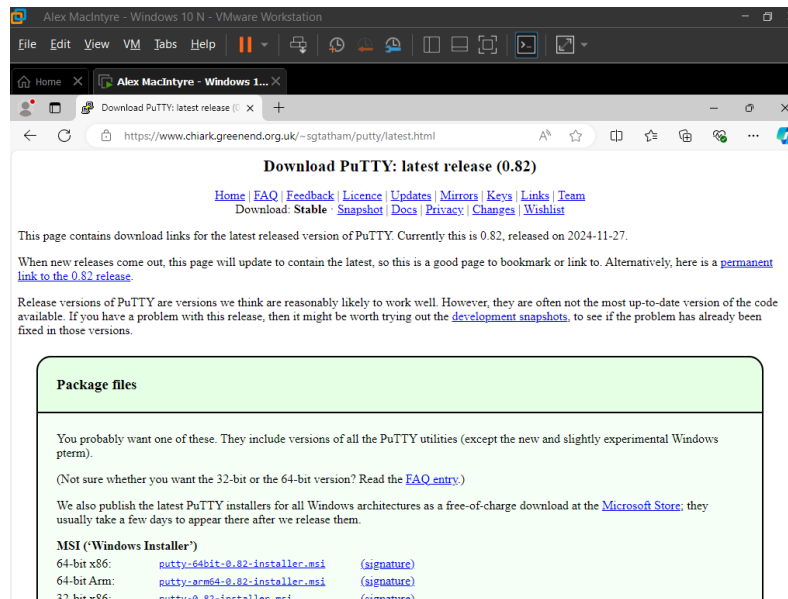

Installing PuTTY on Windows

Step One – Open a web browser and enter the following address:

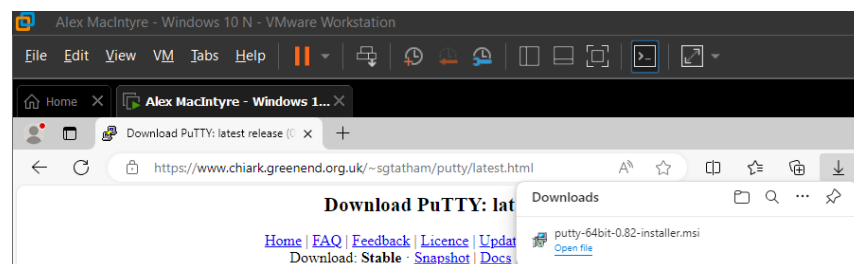
<https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>



Step Two – Click on the link that matches the system’s architecture. In this case, it would be “64-bit x86”.



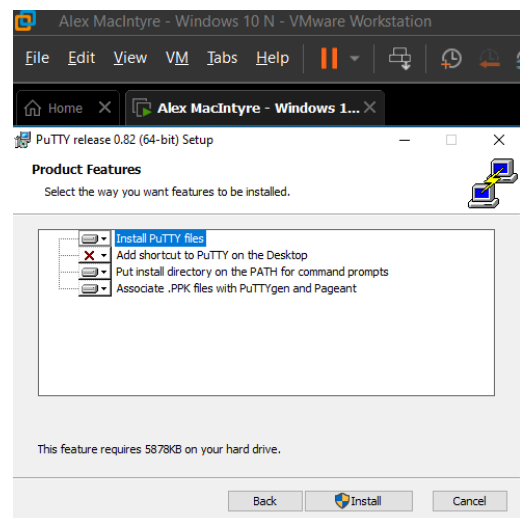
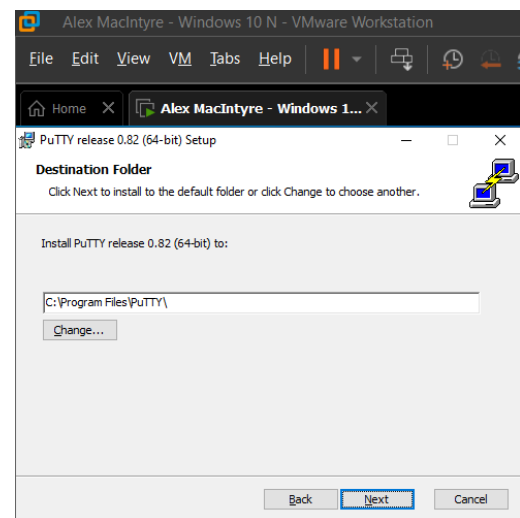
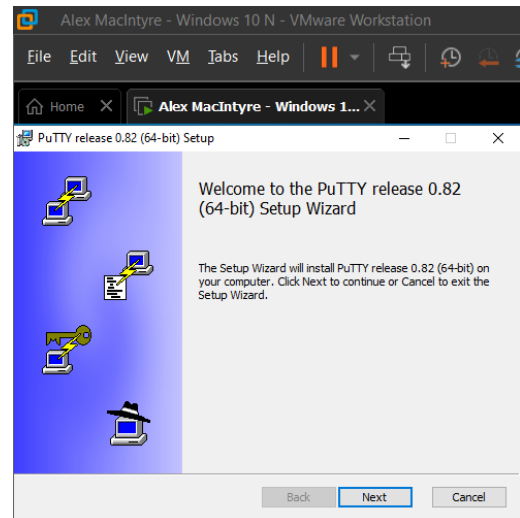
Step Three – Click “Open File” once the download has been completed to begin the installation process.



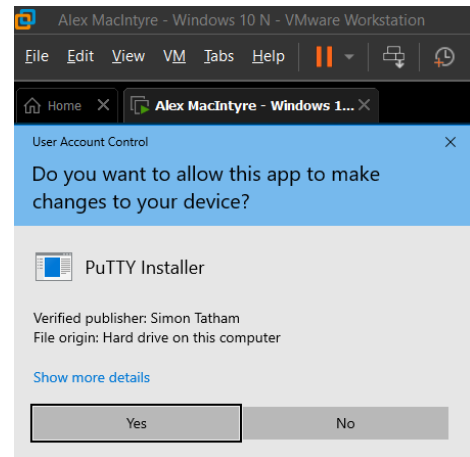
Step Four – Click “Next”.

Step Five – Click “Next”.

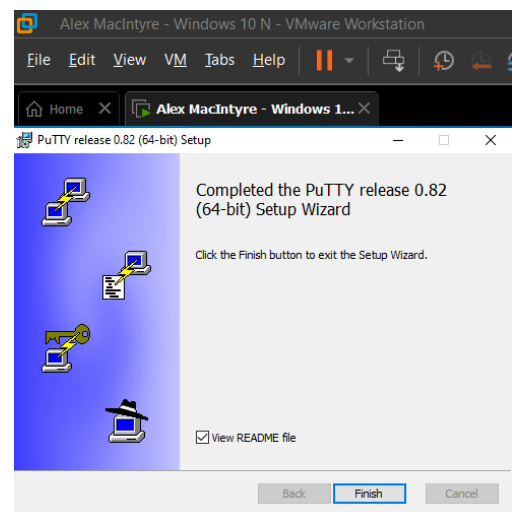
Step Six – Click “Install”.



Step Seven – Allow the installer to make changes.
Click “Yes”.

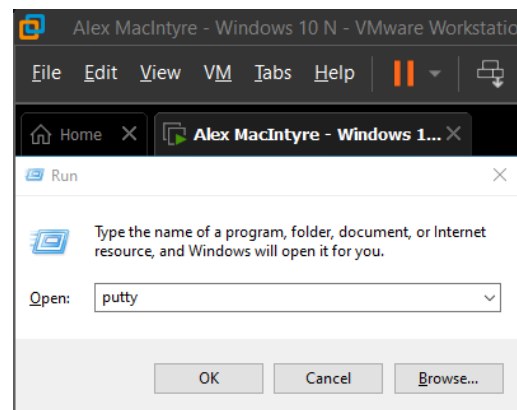


Step Eight – Click “Finish”.

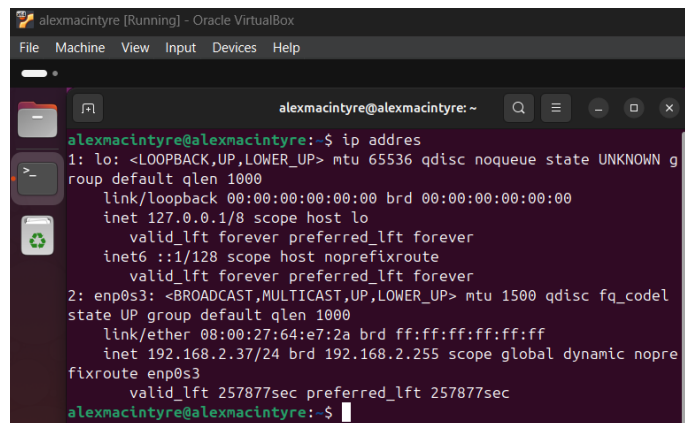


Creating a Connection

Step One – Open PuTTY on Windows. It can be done by pressing **Windows + R**, typing “PuTTY,” and pressing “OK”. This can also be done by pressing the **Windows** key and searching for “PuTTY.”

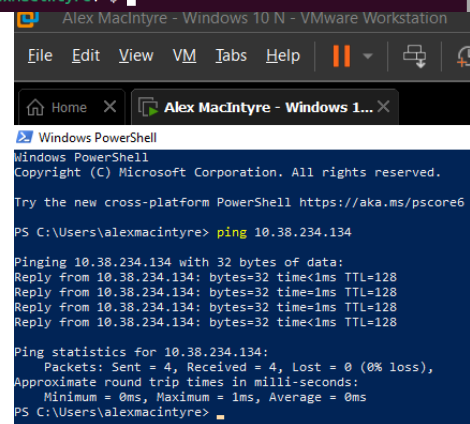


Step Two – In the Ubuntu machine, verify the IP address by typing “**ip address**” into the terminal and pressing Enter. The correct IP address will be under the section beginning with “**2: enp0s3**” and in the “**inet**” line. Here, the IP address is “**192.168.2.37/24**”.



```
alexmacintyre [Running] - Oracle VirtualBox
alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN g
roup default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel
state UP group default qlen 1000
    link/ether 08:00:27:64:e7:2a brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.37/24 brd 192.168.2.255 scope global dynamic nopre
fixroute enp0s3
        valid_lft 257877sec preferred_lft 257877sec
alexmacintyre@alexmacintyre:~$
```

Step Three – On the Windows machine, open PowerShell by pressing “**Windows + X**” and then “**A.**” Ping the Ubuntu machine by typing “**ping**” followed by the IP address assigned to the other ma. Here, the command is “**ping 192.168.2.37**”.



```
Alex MacIntyre - Windows 10 N - VMware Workstation
File Edit View VM Tabs Help
Home X Alex MacIntyre - Windows 1... X
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

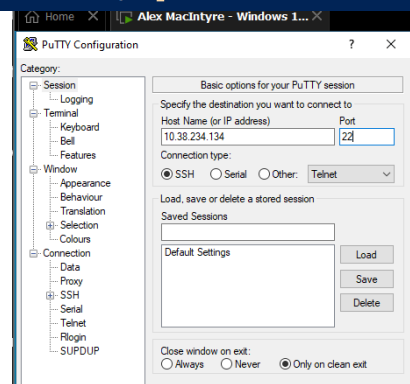
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\alexmacintyre> ping 10.38.234.134

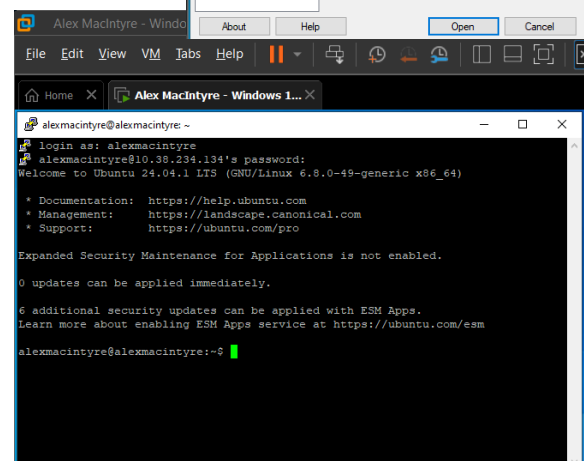
Pinging 10.38.234.134 with 32 bytes of data:
Reply from 10.38.234.134: bytes=32 time<1ms TTL=128
Reply from 10.38.234.134: bytes=32 time<1ms TTL=128
Reply from 10.38.234.134: bytes=32 time<1ms TTL=128
Reply from 10.38.234.134: bytes=32 time<1ms TTL=128

Ping statistics for 10.38.234.134:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\alexmacintyre>
```

Step Four – Now that the Ubuntu IP has been pinged, and it is known here as “**192.168.2.37**,” it can be entered into the “**Host Name (or IP address)**” box in the PuTTY Configuration. Press “**Open**” to begin the connection.



Step Five – Once opened, the terminal will ask for the **sudo username and password**. Input them, and press **Enter** each time. The connection between the two PCs will now be complete.



```
Alex MacIntyre - Windows 10 N - VMware Workstation
File Edit View VM Tabs Help
Home X Alex MacIntyre - Windows 1... X
alexmacintyre@alexmacintyre: ~
login as: alexmacintyre
alexmacintyre@10.38.234.134's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-49-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

alexmacintyre@alexmacintyre:~$
```

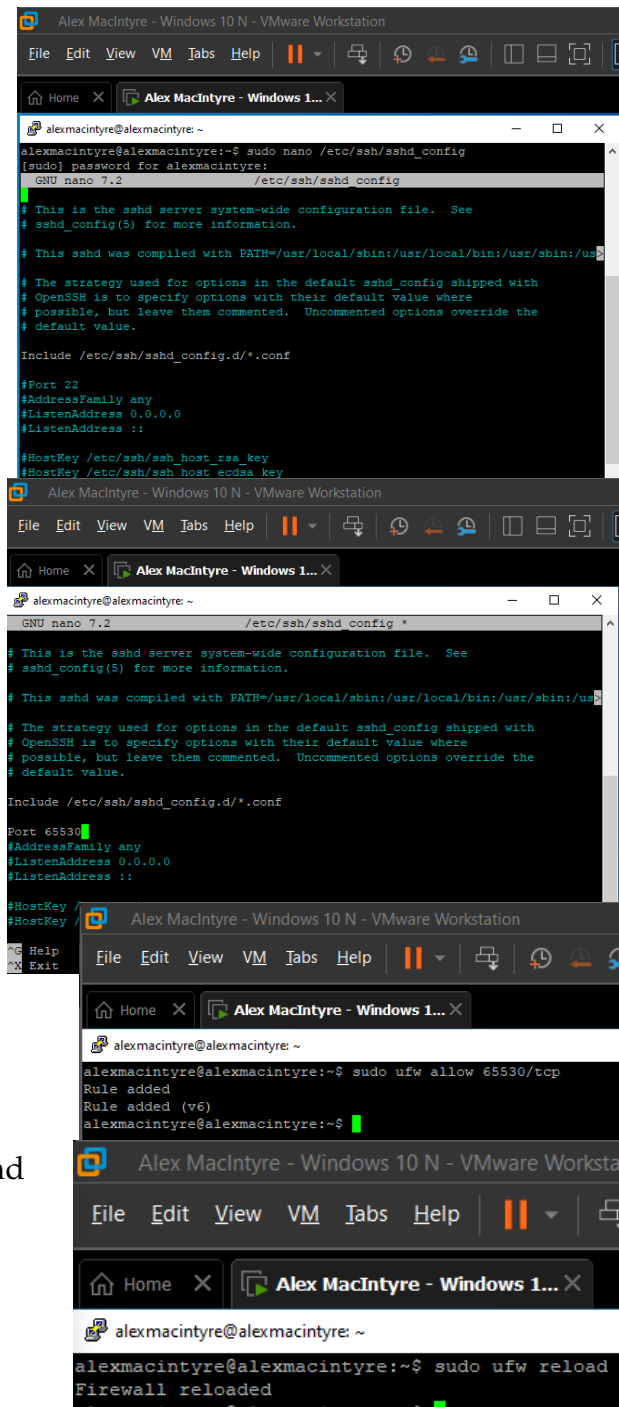
Changing the SSH Port

Step One – In the PuTTY terminal, type “**sudo nano /etc/ssh/sshd_config**” and press **Enter**. Input the **sudo** password and press **Enter** again.

Step Two – To change the port, remove the “#” comment indicator from the start of the line that says “**#Port 22**”. Change the “**22**” to any port number that isn’t identified in a “**ss -ant**” scan, as shown in **Part I – Step Twelve**. Here, it will be “**65530**”. Press “**CTRL+X**” to exit the Text Editor and “**Y**” to save the modification.

Step Three – Type “**sudo ufw allow 65530/tcp**” to allow the new port through the firewall. Press **Enter**.

Step Four – Type “**sudo ufw reload**” to reload the firewall. Press **Enter**.



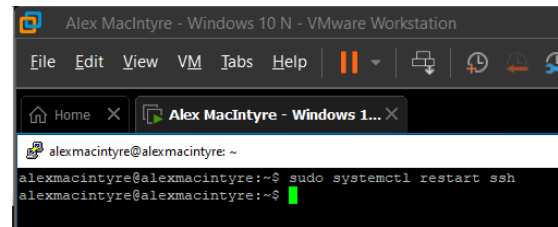
The image displays three sequential screenshots of a terminal window within a VMware Workstation environment, showing the process of changing the SSH port on a system named 'Alex MacIntyre'.

First Screenshot: The terminal shows the command `alexmacintyre@alexmacintyre:~$ sudo nano /etc/ssh/sshd_config` being executed. The user is prompted for the sudo password. The nano editor opens the `/etc/ssh/sshd_config` file. The line `#Port 22` is visible, and the user has changed the port number to 65530, resulting in `Port 65530`.

Second Screenshot: The terminal shows the command `alexmacintyre@alexmacintyre:~$ sudo ufw allow 65530/tcp` being executed. The output shows the rule being added successfully: `Rule added` and `Rule added (v6)`.

Third Screenshot: The terminal shows the command `alexmacintyre@alexmacintyre:~$ sudo ufw reload` being executed. The output shows the firewall being reloaded: `Firewall reloaded`.

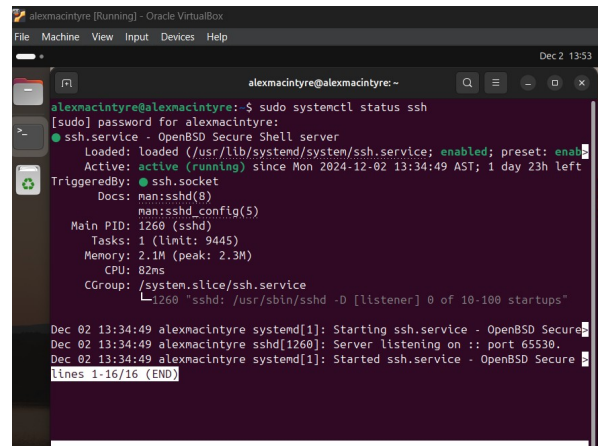
Step Five – For these changes to reflect, type “**sudo systemctl restart ssh**” and press **Enter**.



```
Alex MacIntyre - Windows 10 N - VMware Workstation
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Home X Alex MacIntyre - Windows 1... X
alexmacintyre@alexmacintyre: ~
alexmacintyre@alexmacintyre:~$ sudo systemctl restart ssh
alexmacintyre@alexmacintyre:~$
```

Step Six - After restarting the SSH service, type “**sudo systemctl status ssh**” and press **Enter** to ensure that the listening port is set to the newly established port. It will appear on the second dated line.

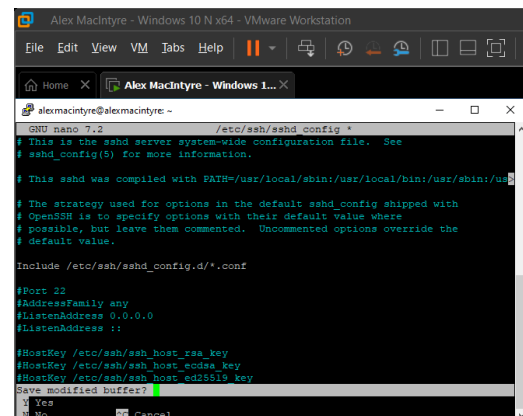
“Server listening on :: **port 65530**”



```
alexmacintyre@alexmacintyre:~$ sudo systemctl status ssh
[sudo] password for alexmacintyre:
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enab
   Active: active (running) since Mon 2024-12-02 13:34:49 AST; 1 day 23h left
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
   Main PID: 1260 (sshd)
   Tasks: 1 (limit: 9445)
   Memory: 2.1M (peak: 2.3M)
   CPU: 82ms
   CGroup: /system.slice/ssh.service
           └─ 1260 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 02 13:34:49 alexmacintyre systemd[1]: Starting ssh.service - OpenBSD Secure
Dec 02 13:34:49 alexmacintyre sshd[1260]: Server listening on :: port 65530.
Dec 02 13:34:49 alexmacintyre systemd[1]: Started ssh.service - OpenBSD Secure
lines 1-16/16 (END)
```

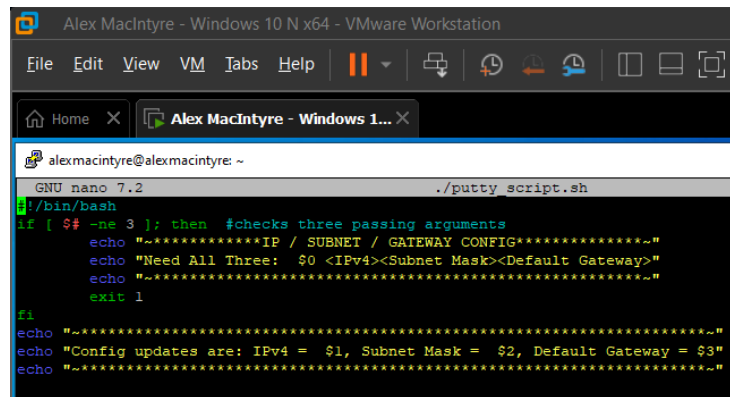
Step Seven – To change the listening port back to Port 22, replace Port 65530 in the text file with Port 22. Press **CTRL+X** and then “**Y**” to save the file. Reload and restart the service with “**sudo ufw reload**” and “**sudo systemctl restart ssh.**”



```
Alex MacIntyre - Windows 10 N x64 - VMware Workstation
File Edit View VM Tabs Help
Home X Alex MacIntyre - Windows 1... X
alexmacintyre@alexmacintyre: ~
GNU nano 7.2 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.
# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/u
# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.
include /etc/ssh/sshd_config.d/*.conf
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::
#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
Save modified buffer
^Y Yes
^N No ^C Cancel
```

Scripting with PuTTY

Script Code



```
alexmacintyre@alexmacintyre: ~  
GNU nano 7.2 ./putty_script.sh  
#!/bin/bash  
if [ $# -ne 3 ]; then #checks three passing arguments  
    echo "~*****IP / SUBNET / GATEWAY CONFIG*****~"  
    echo "Need All Three: $0 <IPv4><Subnet Mask><Default Gateway>"  
    echo "~*****~"  
    exit 1  
fi  
echo "~*****~"  
echo "Config updates are: IPv4 = $1, Subnet Mask = $2, Default Gateway = $3"  
echo "~*****~"
```

Syntax:

if [\$# -ne 3]; then #checks three passing arguments

echo "~*****IP / SUBNET / GATEWAY CONFIG*****~"

echo "Need All Three: \$0 <IPv4><Subnet Mask><Default Gateway>"

echo "~*****~"

exit 1

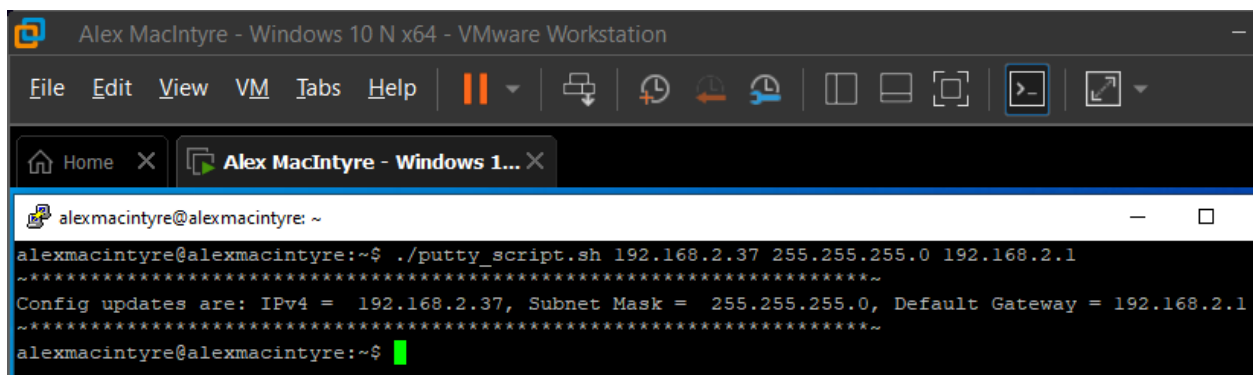
fi

echo "~*****~"

echo "Config updates are: IPv4 = \$1, Subnet Mask = \$2, Default Gateway = \$3"

echo "~*****~"

Working Script



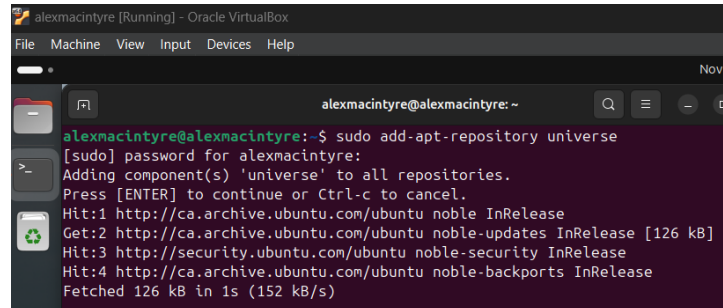
```
alexmacintyre@alexmacintyre: ~  
alexmacintyre@alexmacintyre:~$ ./putty_script.sh 192.168.2.37 255.255.255.0 192.168.2.1  
~*****~  
Config updates are: IPv4 = 192.168.2.37, Subnet Mask = 255.255.255.0, Default Gateway = 192.168.2.1  
~*****~  
alexmacintyre@alexmacintyre:~$
```

**

To show that it was completed successfully, here is a PuTTY installation process on Ubuntu:

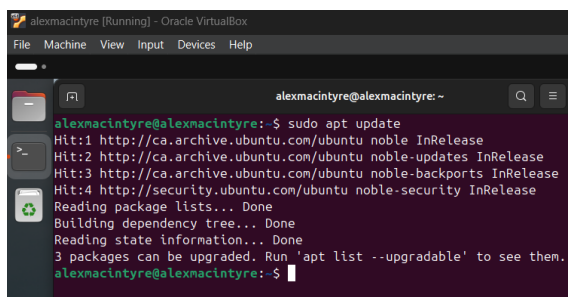
Installing PuTTY on Linux

Step One – To install PuTTY, the Universe Repository will need to be added. Type “**sudo add-apt-repository universe**” and press **Enter**. There will be a prompt to continue. To continue, press **Enter**.



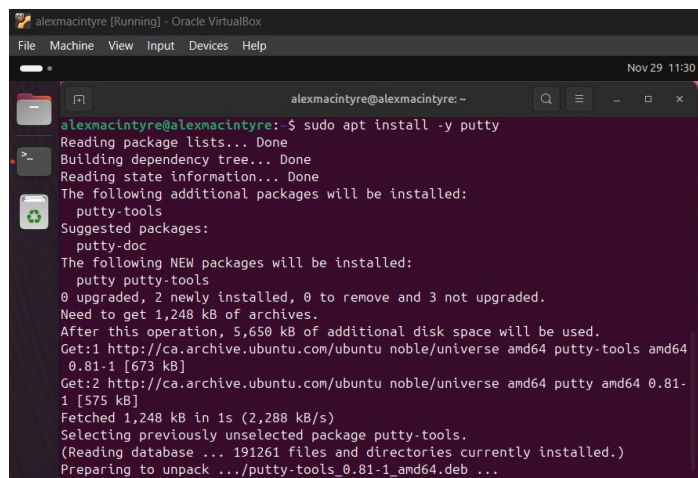
```
alexmacintyre@alexmacintyre:~$ sudo add-apt-repository universe
[sudo] password for alexmacintyre:
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ca.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 1s (152 kB/s)
```

Step Two – Since a new repository was added, there will need to be an update. Type “**sudo apt update**” and press **Enter**.



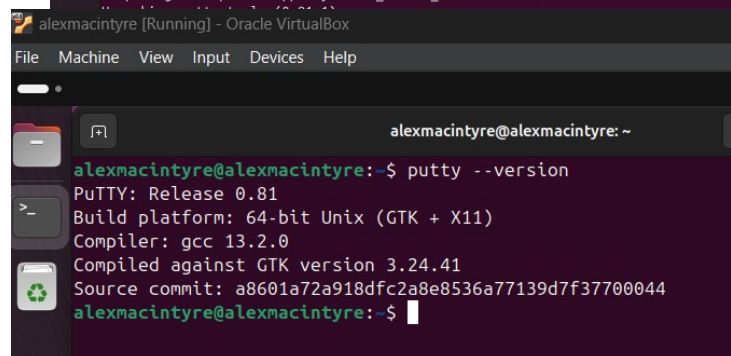
```
alexmacintyre@alexmacintyre:~$ sudo apt update
Hit:1 http://ca.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ca.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ca.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
3 packages can be upgraded. Run 'apt list --upgradable' to see them.
alexmacintyre@alexmacintyre:~$
```

Step Three – Install PuTTY by typing “**sudo apt install -y putty**” and pressing **Enter**.



```
alexmacintyre@alexmacintyre:~$ sudo apt install -y putty
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  putty-tools
Suggested packages:
  putty-doc
The following NEW packages will be installed:
  putty putty-tools
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 1,248 kB of archives.
After this operation, 5,650 kB of additional disk space will be used.
Get:1 http://ca.archive.ubuntu.com/ubuntu noble/universe amd64 putty-tools amd64 0.81-1 [673 kB]
Get:2 http://ca.archive.ubuntu.com/ubuntu noble/universe amd64 putty amd64 0.81-1 [575 kB]
Fetched 1,248 kB in 1s (2,288 kB/s)
Selecting previously unselected package putty-tools.
(Reading database ... 191261 files and directories currently installed.)
Preparing to unpack .../putty-tools_0.81-1_amd64.deb ...
```

Step Four – Verify that PuTTY was successfully installed by typing “**putty --version**” and pressing **Enter**.



```
alexmacintyre@alexmacintyre:~$ putty --version
PuTTY: Release 0.81
Build platform: 64-bit Unix (GTK + X11)
Compiler: gcc 13.2.0
Compiled against GTK version 3.24.41
Source commit: a8601a72a918dfc2a8e8536a77139d7f37700044
alexmacintyre@alexmacintyre:~$
```

**